

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :Electrical Engineering Sem : Vth Subject Name: Advances in Renewable Energy Sources Subject Code: BTEEE505-IV(B) Max Marks: 60 Date: 19/01/2023 Duration: 3 Hr.			
Instructions to the Students: 1. Each question carries 12 marks. 2. Attempt any five questions of the following. 3. Illustrate our answers with neat sketches, diagram etc., wherever necessary. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Classify of biogas plants and their Operating systems.	Analysis	6
B)	What is thermal gasification of biomass? Explain the basic chemistry and applications of a gasifier?	understand	6
C)	Explain Biomass source of Energy and methods of obtaining energy from biomass.	Knowledge	6
Q.2	Solve Two of the following.		
A)	What are the different types of Geothermal sources? Explain two of them.	Knowledge	6
B)	Explain briefly with neat diagram Interconnection of geothermal fossil systems.	understand	6
Q. 3	Solve Any Two of the following.		
A)	What is the Need for hybrid systems? List out different types of hybrid systems.	Knowledge	6
B)	Explain Biogas and solar energy hybrid system with neat diagram.	Application	6
C)	Explain PV–Diesel and battery systems with neat diagram.	Application	6
Q.4	Solve Two of the following.		
A)	Explain effects of CO, NO, CH and particulates in air pollution.	Synthesis	6
B)	Explain causes and effects of acid rain.	Application	6
C)	What are the causes and effects of nuclear pollutions	Analysis	6
Q. 5	Solve Two of the following.		
A)	Describe Environment impact assessment policies and auditing.	Analysis	6
B)	How will you describe the product life cycle assessment and the environmental impact?	Evaluation	6

Q. 6	Solve Two of the following.		
A)	Short note on greenhouse effect.	understand	6
B)	What are the different atmosphere layers? Explain.	Analysis	6
	*** End ***		

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